

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's radius?

Answer

- A. $\sqrt{109}$
- B. $\sqrt{149}$
- C. $\sqrt{167}$
- D. $\sqrt{341}$

Correct Answer: C

Rationale

Choice C is correct. It's given that in the xy -plane, the graph of the given equation is a circle. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the length of the circle's radius. Subtracting $6y$ from both sides of the equation $x^2 + 14x + y^2 = 6y + 109$ yields $x^2 + 14x + y^2 - 6y = 109$. By completing the square, this equation can be rewritten as $(x^2 + 14x + (\frac{14}{2})^2) + (y^2 - 6y + (\frac{-6}{2})^2) = 109 + (\frac{14}{2})^2 + (\frac{-6}{2})^2$. This equation can be rewritten as $(x^2 + 14x + 49) + (y^2 - 6y + 9) = 109 + 49 + 9$, or $(x + 7)^2 + (y - 3)^2 = 167$. Therefore, $r^2 = 167$. Taking the square root of both sides of this equation yields $r = \sqrt{167}$ and $r = -\sqrt{167}$. Since r is the length of the circle's radius, r must be positive. Therefore, the length of the circle's radius is $\sqrt{167}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.